

great and the wave may penetrate only a very short distance before being reduced to a negligibly small percentage of its original strength.

9.9.1 Definition of Depth of Penetration :

The term depth of penetration is defined as that depth in which the wave has been attenuated to $1/e$ i.e. approximately 37% of its original value. The depth of penetration is also called as skin depth.

Consider a wave that is transmitted into the conducting medium. Let $x = 0$ is the boundary of conducting medium with x increasing positively into the conducting medium as shown in Fig. 9.9.1.

We have, $E_y = E_{y0} e^{-\gamma x}$

i.e. $E_y = E_0 \underbrace{e^{-\alpha x}}_{\text{Amplitude}} \times \underbrace{e^{-j\beta x}}_{\text{Phase}}$

Amplitude Phase

It gives variation of E_y in both magnitude and phase as a function of x . The $e^{-\alpha x}$ term indicates attenuation. Thus, when

$$x = 0,$$

$$E_y = E_{y0}$$

$$x = x_1 = 1/\alpha,$$

$$E_y = (E_{y0}/e)$$

and at i.e. we can say that at $\alpha x_1 = 1$, the wave amplitude is $1/e$ times its value at $x = 0$. Thus, by definition, x_1 is the depth of penetration, denoted by δ .

i.e. $\alpha\delta = 1$ or $\delta = 1/\alpha$

Thus, the general expression for depth of penetration using Equation (9.5.7) is

$$\delta = \frac{1}{\alpha} = \frac{1}{\omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \frac{\sigma^2}{\omega^2 \epsilon^2}} - 1 \right)}} \quad \dots(9.9.1)$$

* For good conductor, using the modified expression of α

$$\delta = \frac{1}{\alpha} \approx \sqrt{\frac{2}{\omega\mu\sigma}} \quad \dots(9.9.2)$$

9.9.2 Solved Examples on Depth of Penetration :

Important Formulae

$$\delta = \frac{1}{\alpha} = \frac{1}{\omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \frac{\sigma^2}{\omega^2 \epsilon^2}} - 1 \right)}}$$

$$\delta = \frac{1}{\alpha} \approx \sqrt{\frac{2}{\omega\mu\sigma}}$$

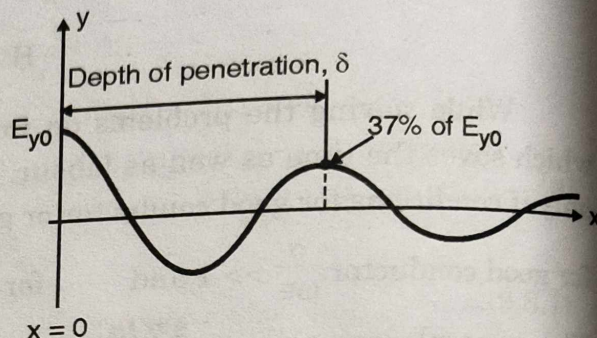


Fig. 9.9.1 : Showing attenuation of a wave in lossy medium

**Significance of phase constant :**

When the wave is propagating its phase is continuously changing indicated by β .

9.6 Mathematical Definition of the Dielectric Media and Conducting Media :

In electromagnetics, materials are divided roughly into two classes

- (1) conductors and (2) dielectrics or insulators.

The dividing line between the two classes is not sharp and some media (the earth for example) are considered as conductors in one part of the radio frequency range, but as dielectric (with loss) in another part of the range.

In Maxwell's first equation : $\nabla \times \bar{H} = \sigma \bar{E} + j\omega\epsilon \bar{E}$

The first term on the RHS is the conduction current density and the second term is the displacement current density. If we take ratio of conduction current density to displacement current density we get, the ratio as,

$$\frac{|J_c|}{|J_d|} = \frac{\sigma}{\omega\epsilon}$$

Naturally

$$\text{For good conductor : } \frac{\sigma}{\omega\epsilon} \gg 1$$

...(9.6.1)

$$\text{For good dielectric : } \frac{\sigma}{\omega\epsilon} \ll 1$$

and $\frac{\sigma}{\omega\epsilon} = 1$ is considered as dividing line between conductor and dielectric.

Computing values of α and β from the given data using Equations (9.5.11) and (9.5.6) definitely is time consuming and labourious. Instead of that I am sure you will be very happy if you have simplified expressions for α and β . The simplification can definitely be done provided that the medium is a good conductor or good dielectrics. Now we will concentrate on modifying Equations (9.5.6) and (9.5.7) as follows :

9.7 Wave Propagation in Good Dielectric :

Using Equation (9.6.1) we can obtain modified expressions for α , β velocity (v), and intrinsic impedance (η) as follows :

9.7.1 Modification of α and β for Good Dielectric :

Using the definition given in Equation (9.6.1), α and β can be modified as given below. As for good dielectric ($\sigma/\omega\epsilon \ll 1$), we can use the approximation as

$$\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} = \left(1 + \frac{\sigma^2}{2\omega^2\epsilon^2}\right) \text{ since for } a \ll 1, \sqrt{1+a^2} = \left(1 + \frac{a^2}{2}\right)$$

Using this in expression of α , we get

$$\alpha \cong \omega \sqrt{\frac{\mu\epsilon}{2} \left(1 + \frac{\sigma^2}{2\omega^2\epsilon^2} - 1\right)}$$

i.e.

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$$

...(9.7.1)

Now for $\beta \cong \omega \sqrt{\frac{\mu\epsilon}{2} \left(1 + \frac{\sigma^2}{2\omega^2\epsilon^2} + 1\right)} = \omega \sqrt{\mu\epsilon \left(1 + \frac{\sigma^2}{4\omega^2\epsilon^2}\right)}$

Now using approximation $\sqrt{1+a^2} = \left(1 + \frac{a^2}{2}\right)$

$$\beta = \omega \sqrt{\mu\epsilon} \left[1 + \frac{\sigma^2}{8\omega^2\epsilon^2}\right]$$

...(9.7.2)

9.7.2 Velocity in Good Dielectric :

Now the velocity of wave, $v = \omega / \beta = \frac{1}{\sqrt{\mu\epsilon} \left[1 + \frac{\sigma^2}{8\omega^2\epsilon^2}\right]}$

Again using approximation for $a \ll 1$ as $\frac{1}{1+a} = (1 - a)$

$$v = \frac{1}{\sqrt{\mu\epsilon}} \left(1 - \frac{\sigma^2}{8\omega^2\epsilon^2}\right)$$

i.e.

$$v = v_0 \left[1 - \frac{\sigma^2}{8\omega^2\epsilon^2}\right]$$

...(9.7.3)

9.7.3 η in Good Dielectric :

Taking $j\omega\epsilon$ common from denominator $\eta = \sqrt{\frac{\mu\epsilon}{[1 + (\sigma / j\omega\epsilon)]}}$... (A)

For a good dielectric, $(\sigma / \omega\epsilon) \ll 1$, Then

$$\eta = \sqrt{\frac{\mu}{\epsilon} \left[1 - \frac{\sigma}{j\omega\epsilon}\right]} \quad \dots \text{using approximation, } \frac{1}{1+a} = 1 - a$$

or

$$\eta = \sqrt{\frac{\mu}{\epsilon} \left[1 + j \frac{\sigma}{2\omega\epsilon}\right]}$$

...(9.7.4)

whenever a small amount of loss is present in the intrinsic impedance a small reactive component gets added.

9.8 Wave Propagation in Good Conductor :

In the previous sections we studied the propagation in good dielectric, free space and lossy dielectric. Its time now study propagation in good conductor.

For good dielectrics we have, $(\sigma / \omega \epsilon) \gg 1$

9.8.1 To find α and β for Good Conductor :

Using expression for γ as

$$\gamma^2 = j\omega\mu [\sigma + j\omega\epsilon] = j\omega\mu\sigma \left[1 + \frac{j\omega\epsilon}{\sigma} \right]$$

$$= j\omega\mu\sigma \quad \text{as } \dots (\sigma/\omega\epsilon) \gg 1$$

...(a)

i.e. $\gamma = \sqrt{j\omega\mu\sigma} = \sqrt{\omega\mu\sigma} \angle 90^\circ$

or $\gamma = \sqrt{\omega\mu\sigma} \angle 45^\circ$

...(9.8.1)

Now $\gamma = \alpha + j\beta$

...(b)

Squaring Equation (b) and equating with Equation (a)

$$\gamma^2 = \alpha^2 - \beta^2 + 2j\alpha\beta = j\omega\mu\sigma$$

comparing real parts $\alpha^2 - \beta^2 = 0$ i.e. $\alpha = \beta$

and comparing imaginary terms

$$2\alpha\beta = \omega\mu\sigma$$

i.e.

$$2\alpha^2 = \omega\mu\sigma$$

$\therefore$

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$$

...(9.8.2)

and

$$\beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

...(9.8.3)

9.8.2 Velocity of Propagation for Good Conductor :

$$v = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\omega\mu\sigma/2}}$$

or

$$v = \sqrt{\frac{2\omega}{\mu\sigma}}$$

...(9.8.4)

9.8.3 Intrinsic Impedance for Good Conductor :

Now for a good conductor, $(\sigma/\omega\epsilon) \gg 1$, then Equation (A) becomes

$$\eta = \sqrt{\frac{\mu}{\epsilon} \times \frac{j\omega\epsilon}{\sigma}} = \sqrt{\frac{j\omega\mu}{\sigma}}$$

...because 1 can be neglected.

$\therefore$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ$$

$$(\text{As } \sqrt{jy} = \sqrt{y} \angle \pi/2 = \sqrt{y} \angle \pi/4)$$

...(9.8.5)

$$\eta = \sqrt{\frac{\omega\mu}{2\sigma}} (1 + j1)$$

Then $\text{Re}(\eta) = \sqrt{\frac{\omega\mu}{2\sigma}}$ and $\text{Im}(\eta) = \sqrt{\frac{\omega\mu}{2\sigma}}$

Note that η has phase angle 45° , which reflects on to the phase difference between E and H

$$E = E_0 e^{-\alpha x} \cos(\omega t - \beta x)$$

$$H = \frac{E_0}{|\eta|} e^{-\alpha x} \cos(\omega t - \beta x - \pi/4)$$

While solving the problems on finding α , β , γ , η we can use simplified expressions which saves the time as well as labour. But be careful because these expressions are used only if conditions for good conductor or good dielectric are satisfied, which are

for good conductor $\frac{\sigma}{\omega\epsilon} \gg 1$ and for good dielectrics $\frac{\sigma}{\omega\epsilon} \ll 1$

The general expressions which we derived earlier as well as simplified expressions for α , β , γ , v and η are given below for fast reference while solving the problems.

General Expressions	
$E_y = E_{y0} e^{-\gamma x}$	$\gamma^2 = j\omega\mu(\sigma + j\omega\epsilon)$
$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} - 1 \right)}$	$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} + 1 \right)}$
$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$	

Simplified Expressions

Example 9.9.1 : A medium has the following parameters :

$$\mu_r = 10, \epsilon_r = 2.5, \sigma = 10^{-4} \text{ mho / m}$$

Determine $\alpha, \beta, \lambda, v, \eta$, for 1 GHz.

MU - Dec. 2002

Solution : If we want to take advantage of simplified expressions for good conductor, check the ratio

$$\frac{\sigma}{\omega\epsilon} = \frac{10^{-4}}{2\pi \times 10^9 \times 2.5 \times 8.854 \times 10^{-12}} = 7.19 \times 10^{-4} \ll 1$$

So at the given frequency the medium is not a good conductor, so use exact expressions.

We have propagation constant

$$\begin{aligned} \gamma &= \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \\ &= \sqrt{j \times 2\pi \times 1 \times 10^9 \times 10 \times 4\pi \times 10^{-7} \times \left(10^{-4} + j \times 2\pi \times 1 \times 10^9 \times 2.5 \times \frac{10^{-9}}{36\pi}\right)} \\ &= \sqrt{j 78956.8 (10^{-4} + j 0.1391)} \\ &= \sqrt{78956.8 \angle 90^\circ (0.1391 + j 89.96^\circ)} = 104.80 \angle 89.98^\circ = 0.0366 + j 104.80 \end{aligned}$$

But $\gamma = \alpha + j\beta \therefore \alpha = 0.0366 \text{ (Np/m)}$ and $\beta = 104.80 \text{ (rad/m)}$

Now, $\lambda = \frac{2\pi}{\beta} = 0.05995 \text{ (m)}$ $v = \frac{\omega}{\beta} = \frac{2\pi \times 1 \times 10^9}{104.80} = 56.95 \times 10^6 \text{ (m/s)}$

$$\begin{aligned} \eta &= \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j \times 2\pi \times 1 \times 10^9 \times 4\pi \times 10^{-7} \times 10}{10^{-4} + j \times 2\pi \times 1 \times 10^9 \times 8.854 \times 10^{-12} \times 2.5}} \\ &= \sqrt{\frac{78956.8 \angle 90^\circ}{0.1391 \angle 89.96^\circ}} = 753.41 \angle 0.02^\circ = 753.41 + j 0.263 \text{ (}\Omega\text{)} \end{aligned}$$

...Ans.

Example 9.9.2 : Calculate intrinsic impedance η , the propagation constant γ , the wave velocity v for a conducting medium in which $\sigma = 60 \text{ Ms/m}$, $\mu_r = 1$, at a frequency $f = 200 \text{ MHz}$.

MU - Dec. 2007

Solution : It is given that the medium is a conductor. Let us check whether it is a good conductor or not.

$$\frac{\sigma}{\omega\epsilon} = \frac{60 \times 10^6}{2\pi \times 200 \times 10^6 \times (10^{-9}/36\pi)} = 5.4 \times 10^9 \gg 1$$

So the given media is a good conductor. Using approximate relations,

$$\begin{aligned} \gamma &= \sqrt{\omega\mu\sigma} \angle 45^\circ = \sqrt{2\pi \times 200 \times 10^6 (4\pi \times 10^{-7}) \times 60 \times 10^6} \angle 45^\circ \\ &= 307.81 \times 10^3 \angle 45^\circ \text{ (m}^{-1}\text{)} \end{aligned}$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ = \sqrt{\frac{2\pi \times 200 \times 10^6 \times 4\pi \times 10^{-7}}{60 \times 10^6}} \angle 45^\circ = 5.130 \times 10^{-3} \angle 45^\circ (\Omega)$$

The calculated γ can be written as, $\gamma = 307.81 \times 10^3 \angle 45^\circ = 2.18 \times 10^5 + j 2.18 \times 10^5$

Equating with $\alpha + j\beta$ we get, $\alpha = \beta = 2.18 \times 10^5$ (Np/m), (rad/m)

The velocity is given by, $v = \frac{\omega}{\beta} = \frac{2\pi \times 200 \times 10^6}{2.18 \times 10^5} = 5.76 \times 10^3$ (m/s) ...Ans.

Example 9.9.3 : Calculate the intrinsic impedance η and propagation constant γ for a conducting medium in which $\sigma = 58 \times 10^6$ (mho/m), $\mu_r = 1$ at a frequency of $f = 100 \times 10^6$ Hz.

Solution : This problem is similar to previous problem, the only change is in the value of σ . So, go on the similar lines, answer is given below.

$$\begin{aligned} \frac{\sigma}{\omega\epsilon} &= 1 \times 10^{12} \gg 1 & \gamma &= 214 \times 10^5 \angle 45^\circ (\text{m}^{-1}) \\ \eta &= 3.69 \times 10^{-3} \angle 45^\circ (\Omega) & \alpha &= \beta = 1.51 \times 10^5 \text{ (Np / m), (rad/m)} \\ v &= 4.16 \times 10^3 \text{ (m/s).} & & \dots\text{Ans.} \end{aligned}$$

Example 9.9.4 : Wet, marshy soil is characterized by $\sigma = 10^{-2}$ S/m, $\epsilon_r = 15$ and $\mu_r = 1$. Show that at a frequency of 60 Hz, it can be considered as good conductor. Hence, calculate at 60 Hz, (i) Skin depth, (ii) Intrinsic impedance, (iii) Propagation constant.

Solution : For the medium to be good conductor we have the condition $(\sigma / \omega\epsilon) \gg 1$. Let us check whether this condition is satisfied for given medium.

$$\frac{\sigma}{\omega\epsilon} = \frac{10^{-2}}{2\pi \times 60 \times \frac{10^{-9}}{36\pi} \times 15} = 200000 \gg 1$$

hence given medium can be considered as good conductor at 60 Hz.

(i) **To calculate Skin depth :**

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{2}{2\pi \times 60 \times 4\pi \times 10^{-7} \times 10^{-2}}} = 649.75 \text{ m} \dots\text{Ans.}$$

(ii) **To calculate Intrinsic impedance :**

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j \times 2\pi \times 60 \times 4\pi \times 10^{-7}}{10^{-2} + j \times 2\pi \times 60 \times \frac{10^{-9}}{36\pi} \times 15}} = 0.217 \angle 45^\circ (\Omega) \dots\text{Ans.}$$

(iii) **To calculate propagation constant :** We have, $\gamma = \alpha + j\beta$

For good conductor, $\alpha = \beta = 1/\delta = 1.539 \times 10^{-3}$

Hence, $\gamma = 1.539 \times 10^{-3} + j 1.539 \times 10^{-3}$

Example 9.9.5 : A plane EM wave in a low loss dielectric has a velocity 3×10^8 (m/s). Calculate the intrinsic impedance and skin depth of copper at a frequency of 100 MHz if the wave is incident on it. Given : $\sigma_{cu} = 58 \times 10^6$ (mho/m).

Solution : This example is similar to previous problem, the only change is in the data given. So go on the similar lines, answers are given below.

$$\frac{\sigma}{\omega\epsilon} = 1 \times 10^{12} \gg 1$$

$$\gamma = 214 \times 10^5 \angle 45^\circ \text{ (m}^{-1}\text{)}$$

$$\eta = 3.69 \times 10^{-3} \angle 45^\circ \text{ (}\Omega\text{)}$$

$$\alpha = 1.51 \times 10^5 \text{ (Np/m)}$$

$$\delta = \frac{1}{\alpha} = 6.62 \times 10^{-6} \text{ (m).}$$

...Ans.

Example 9.9.6 : Find the skin depth at frequency of 1.6 MHz in aluminium whose $\sigma = 38.2 \text{ MS / m}$ and $\mu_r = 1.0$. Also find propagation constant and wave velocity propagating in this medium. (assume $\alpha = \beta = 1/\delta$).

MU - Dec. 2009, 10 Marks

Solution : We have skin depth,

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{2}{2\pi \times 1.6 \times 10^6 \times 1.0 \times 4\pi \times 10^{-7} \times 38.2 \times 10^6}} = 64.73 \text{ } \mu\text{m}$$

$$\therefore \alpha = \frac{1}{\delta} = 15.45 \times 10^3 \text{ (Np/m)}$$

The propagation constant is, $\gamma = \alpha + j\beta$

For good conductor, $\alpha = \beta$

Thus, propagation constant, $\gamma = 15.45 \times 10^3 + j15.45 \times 10^3$

We have the expression for wave velocity as $v = \frac{\omega}{\beta} = \frac{2\pi \times 1.6 \times 10^6}{15.45 \times 10^3} = 651.53 \text{ (m/s)}$

Example 9.9.7 : What is the skin depth of current penetration in copper at a frequency of 10^4 MHz, if the resistivity is $1.7 \times 10^{-6} \text{ ohm-cm}$.

Solution : Given, resistivity = $1.7 \times 10^{-6} \text{ } \Omega\text{-cm} = 1.7 \times 10^{-8} \text{ } \Omega\text{-m}$

Conductivity of copper is,

$$\sigma = \frac{1}{\text{Resistivity}} = \frac{1}{1.7 \times 10^{-8}} = 58.82 \text{ Mmho / m.}$$

To check whether copper is a good conductor at operating frequency

$$\frac{\sigma}{\omega\epsilon} = \frac{58.82 \times 10^6}{2\pi \times 10^4 \times 10^6 \times \frac{10^{-9}}{36\pi}} = 0.1 \times 10^9 \gg 1$$

Thus copper is a good conductor. The skin depth for good conductor is,

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{2}{2\pi \times 10^4 \times 10^6 \times 4\pi \times 10^{-7} \times 58.82 \times 10^6}} = 0.656 \text{ } \mu\text{m} \quad \text{...Ans.}$$



Example 9.9.8 : Find the depth of penetration if for copper $\mu_r = 1$, $\sigma = 58 \text{ Mmho/m}$ at different frequencies $f = 60 \text{ Hz}$, 1 MHz and 30 GHz .

Solution : First of all check whether Cu is a good conductor at the highest frequency,

$$\text{i.e. } \frac{\sigma}{\omega\epsilon} = \frac{58 \times 10^6}{2\pi \times 30 \times 10^9 \times (10^{-9}/36\pi)} = 34.8 \times 10^5 \geq 1$$

Thus, Cu is a good conductor. Using the expression for δ for good conductor.

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{2}{2\pi f \times 4\pi \times 10^{-7} \times 58 \times 10^6}} = \frac{66 \times 10^{-3}}{\sqrt{f}}$$

$$\text{at } f = 60 \text{ Hz} \quad \delta = 8.5 \times 10^{-3} \text{ m}$$

$$f = 1 \text{ MHz} \quad \delta = 6.6 \times 10^{-5} \text{ m}$$

$$f = 30 \times 10^9 \quad \delta = 3.8 \times 10^{-7} \text{ m}$$

...Ans.

Example 9.9.9 : At what frequency may earth be considered a perfect dielectric if $\sigma = 5 \times 10^{-3} \text{ (S/m)}$, $\mu_r = 1$ and $\epsilon_r = 8$. Can α be assumed zero at these frequencies.

Solution : For dielectric $(\sigma / \omega\epsilon) \ll 1$.

If we assume $\ll 1$ means less than 0.01, then using the limiting value,

$$\text{i.e. } \frac{\sigma}{\omega\epsilon} = 0.01 \therefore \omega = \frac{\sigma}{0.01\epsilon} = \frac{100 \sigma}{\epsilon}$$

$$f = \frac{\omega}{2\pi} = \frac{100 \sigma}{2\pi \times \epsilon} = \frac{100 \times 5 \times 10^{-3}}{2\pi \times (10^{-9}/36\pi) \times 8} = 1.125 \times 10^9 \text{ (Hz)} = 1.125 \text{ (GHz)}$$

Now for good dielectric we have

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} = \frac{5 \times 10^{-3}}{2} \sqrt{\frac{4\pi \times 10^{-7} \times 1}{(10^{-9}/36\pi) \times 8}} = 0.333 \text{ Np/m.} \quad \dots\text{Ans.}$$

Thus, α can not be assumed zero.

Example 9.9.10 : For sea water with $\sigma = 5 \text{ mho/m}$ and $\epsilon_r = 80$, $\mu = \mu_0$, find the distance a radio signal can be transmitted at 25 Kcps and 25 Mcps . If the range is taken to be the distance at which 90% of the wave amplitude is attenuated.

Solution : We know that the electric field intensity is given by

$$E = \underbrace{E_0 e^{-\alpha x}}_{\text{amplitude}} \times \underbrace{e^{-j\beta x}}_{\text{phase}}$$

As phase angle is not important, $E = E_0 e^{-\alpha x}$

Let $x = x_1$ be the distance at which E drops to $0.1 E_0$

Poynting Theorem

10.1 Introduction :

In the transmitter the wave is generated by a source, then it travels to a distant receiving point. Whenever a wave travels from one point to other, transfer of energy takes place. There is a simple and direct relation between the rate of the energy transfer and the amplitudes of electric and magnetic intensities associated with a travelling electromagnetic wave. This relation is known as Poynting vector after I.H. Poynting. Many of us think of the Poynting vector as a "Pointing" vector. This homonym, while accidental, is correct.

10.2 Poynting Theorem :

MU - Dec. 2004, Dec. 2005, Dec. 2006, May 2006, May 2007, May 2008, Dec. 2008, May 2009, Dec. 2009

10.2.1 Statement of Poynting Theorem :

Vector product of electric intensity $\vec{E}$ and magnetic intensity $\vec{H}$ at any point is a measure of the rate of energy flow per unit area at that point.

10.2.2 Mathematical Expression of Poynting Theorem :

Mathematically the theorem can be expressed as

$$\vec{P} = \vec{E} \times \vec{H}$$

The direction of flow is perpendicular to $\vec{E}$ and $\vec{H}$ in the direction of the vector $\vec{E} \times \vec{H}$. The relation can be obtained from Maxwell's equations as follows. The magnetomotive force equation is,

$$\nabla \times \vec{H} = \vec{J} + \dot{\vec{D}} \quad \dots (1)$$

or

$$\vec{J} = \nabla \times \vec{H} - \dot{\vec{D}}$$

In order to convert $\vec{J}$ from the dimension of current density to power per unit volume,



multiply both sides of the above expression by $\vec{E}$ we have,

$$\vec{E} \cdot \vec{J} = \vec{E} \cdot (\nabla \times \vec{H}) - \epsilon \vec{E} \cdot \dot{\vec{E}} \quad \dots (i)$$

We have vector identity

$$\nabla \cdot (\vec{P} \times \vec{Q}) = \vec{Q} \cdot (\nabla \times \vec{P}) - \vec{P} \cdot (\nabla \times \vec{Q})$$

where, $\vec{P}$ and $\vec{Q}$ are any vector functions. Replacing $\vec{P}$ by $\vec{E}$ and $\vec{Q}$ by $\vec{H}$,

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H}).$$

Putting the value of $\vec{E} \cdot (\nabla \times \vec{H})$ from above equation in Equation (i),

$$\vec{E} \cdot \vec{J} = \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H}) - \epsilon \vec{E} \cdot \dot{\vec{E}}$$

Using the second Maxwell's equation $\nabla \times \vec{E} = -\mu \dot{\vec{H}}$

in the expression for $\vec{E} \cdot \vec{J}$ we get $\vec{E} \cdot \vec{J} = -\mu \vec{H} \cdot \dot{\vec{H}} - \epsilon \vec{E} \cdot \dot{\vec{E}} - \nabla \cdot (\vec{E} \times \vec{H})$

Now,

$$\vec{H} \cdot \dot{\vec{H}} = \frac{1}{2} \frac{\partial}{\partial t} H^2 \quad \text{and} \quad \vec{E} \cdot \dot{\vec{E}} = \frac{1}{2} \frac{\partial}{\partial t} E^2$$

So that

$$\vec{E} \cdot \vec{J} = -\frac{\mu}{2} \frac{\partial}{\partial t} H^2 - \frac{\epsilon}{2} \frac{\partial}{\partial t} E^2 - \nabla \cdot (\vec{E} \times \vec{H})$$

Integrating over a volume v , $\int_v (\vec{E} \cdot \vec{J}) dv = -\frac{\partial}{\partial t} \int_v \left(\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right) dv - \int_v \nabla \cdot (\vec{E} \times \vec{H}) dv \dots (ii)$

Using the divergence theorem the last term can be changed from volume integral to surface integral,

$$\int_v \nabla \cdot (\vec{E} \times \vec{H}) dv = \oint_s (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

Then Equation (ii) becomes

$$\int_v (\vec{E} \cdot \vec{J}) dv = -\frac{\partial}{\partial t} \int_v \left(\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right) dv - \oint_s (\vec{E} \times \vec{H}) \cdot d\vec{s} \quad \dots (10.2.1)$$

Equation (10.2.1) shows that the instantaneous power flow contains stored energy and the total power flowing in the volume.

10.2.3 A Physical Interpretation of Poynting Vector :

A physical interpretation of the above equation leads to some interesting conclusions :

- (i) The term $\int_v (\vec{E} \cdot \vec{J}) dv$ on the left hand side represents the (instantaneous) power dissipated in the volume v . This result is obtained as a generalization of Joule's law. A conductor of cross-sectional area A , carrying a

current I and having a voltage drop E per unit length (V/m) will have a power loss of EI watt per unit length. The power dissipated per unit volume will be $EI/A = EJ$ watt per unit volume. In this case V and $\bar{J}$ are in the same direction. In general where this may not be true, the power dissipated per unit volume would still be given by the product of $\bar{J}$ and the component of $\bar{E}$ having the same direction as $\bar{J}$ or $\bar{E} \cdot \bar{J}$. The total power dissipated in the volume v would be

$$\int_v (\bar{E} \cdot \bar{J}) dv \quad \dots(iii)$$

When E in this expression represents the electric intensity required to produce the current density $\bar{J}$ in the conducting medium, the Equation (iii) represents power dissipated as ohmic (I^2R) loss. However, if $\bar{E}$ is an electric intensity due to a source of power, e.g., due to a battery, then the power represented by Equation (iii) would be used up in driving the current against the battery voltage and hence charging the battery. If the direction of $\bar{E}$ were opposite to that $\bar{J}$ the dissipated power represented by Equation (iii) would be negative. In this case, the battery would be generating electric power.

- (ii) The first term $-\frac{\partial}{\partial t} \int_v \left(\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right) dv$ on the right hand side of

Equation (10.2.1) without the negative time derivative represents the stored (electric and magnetic) energy in the volume. The negative time derivative of this quantity then represents the rate at which the stored energy in the volume is decreasing.

- (iii) The interpretation of the remaining term $-\oint_s (\bar{E} \times \bar{H}) \cdot d\bar{s}$ follows from the

application of the law of conservation of energy. The rate of energy dissipation in the volume v must equal the rate at which the stored energy in v is decreasing plus the rate at which energy is entering the volume v from outside. The term

$$-\oint_s (\bar{E} \times \bar{H}) \cdot d\bar{s}$$

therefore represents the rate of flow of energy inward through the surface of the volume. Then this expression without the negative sign, i.e.,



$$\int_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

represents the **rate of flow of energy outward through the surface enclosing the volume**. Thus, the integral of $\vec{E} \times \vec{H}$ over any surface gives the rate of energy flow through that surface. It is seen that the vector

$$\vec{P} = \vec{E} \times \vec{H}$$

... (10.2.2)

has the dimensions of watt / sq.m.

Equation (10.2.2) is the expression Poynting vector of $\vec{P}$.

10.2.4 Significance of Poynting Vector :

In Poynting theorem the vector product $\vec{P} = \vec{E} \times \vec{H}$ at any point is a measure of the rate of energy flow per unit area at that point. The direction of flow is perpendicular to $\vec{E}$ and $\vec{H}$ in the direction of the vector $\vec{E} \times \vec{H}$. The direction of the vector $\vec{P}$ indicates the direction of the instantaneous power flow at the point, and many of us think of the Poynting vector as a "Pointing" vector. This homonym, while accidental, is correct.

Note : Poynting vector physically denotes the power density leaving the given volume in time varying field.

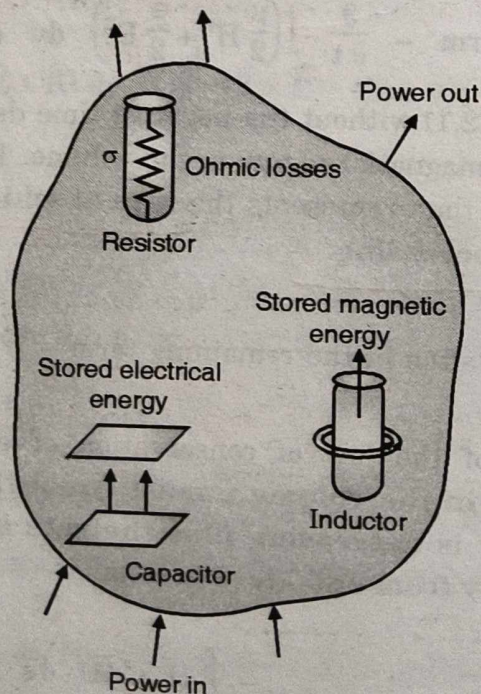


Fig. 10.2.1 : Power balance for EM fields

The energy stored in the system is represented by energy storing elements like capacitor and inductor. The power dissipated is accounted by resistor. The diagrammatic representation is shown in Fig. 10.2.1.

From the physical interpretation now it is clear that the Poynting theorem relates, the incoming energy, outgoing energy, energy stored and dissipated in the system. In other words the **"Poynting theorem gives the energy balance in the system or it proves law of conservation of energy"**.

10.3 Power Flow for a Plane Wave :

We know that energy density due to electric and magnetic fields is given by

$$\frac{1}{2} (\epsilon E^2 + \mu H^2)$$

When a wave is travelling it travels with a velocity $v_0 = \frac{1}{\sqrt{\mu \epsilon}}$

For a wave moving with a velocity v_0 the rate of flow of energy per unit area would be

$$P = \frac{1}{2} (\epsilon E^2 + \mu H^2) v_0$$

The electric field (E) and magnetic field (H) for a plane wave are related by

$$\frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}}$$

Using this equation, the equation previous to it can be written as

$$P = \frac{1}{2} \left(\epsilon \sqrt{\frac{\mu}{\epsilon}} EH + \mu \sqrt{\frac{\epsilon}{\mu}} EH \right) v_0 = \left(\frac{EH}{v_0} \right) v_0 = \bar{\mathbf{E}} \times \bar{\mathbf{H}}$$

This gives us power flow for a plane wave.

10.4 Power Flow in a Concentric Cable :

Consider a concentric cable with conductors having negligible resistance. The radius of the inner conductor is 'a' and the inside radius of the out conductor is 'b'. This cable is used to transfer power to a load resistance R along a cable. The voltage between conductors is V and a steady current I is flowing in the inner and outer conductor.

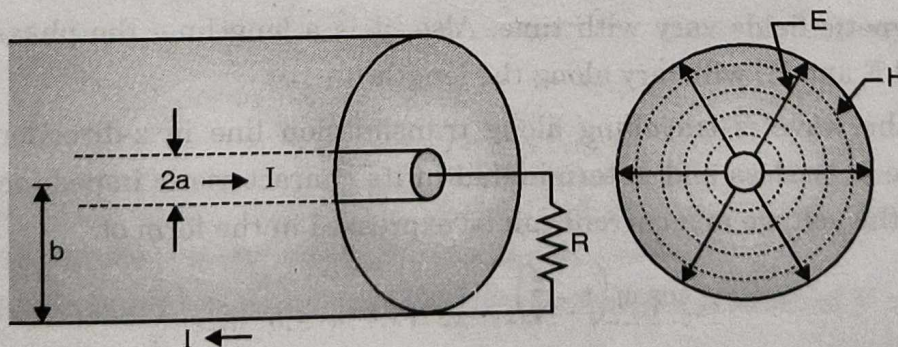


Fig. 10.4.1 : Power flow in a cable

At every point in a plane parallel to the x and y axes, $\vec{P}$ will be parallel to the z -axis. The Poynting vector will be given by

$$\vec{P} = \vec{E} \times \vec{H} = A \cos^2 \omega \left(t - \frac{z}{v} \right) f(x, y)$$

The function $f(x, y)$ will not vary with z or t , it is a function of x and y only.

The expression for $\vec{P}$ says that for a fixed value of time, the power passing through a plane will vary with z i.e. the position of the plane. Whereas for a fixed z , the power will vary with time.

10.6 Instantaneous, Average and Complex Poynting Vector :

We know that in an AC circuit, the instantaneous power $\tilde{W}$ is always given by the product of the instantaneous voltage $\tilde{V}$ and the instantaneous current I .

$$\tilde{W} = \tilde{V} \tilde{I}$$

The quantities $\tilde{V}$ and $\tilde{I}$ are obtained by multiplying V and I by $e^{j\omega t}$ and taking real part i.e.

$$\tilde{V} = \text{Re} \{ V e^{j\omega t} \} = \text{Re} \{ |V| e^{j\theta_v} e^{j\omega t} \} = |V| \cos(\omega t + \theta_v) \quad \dots(i)$$

$$\tilde{I} = \text{Re} \{ I e^{j\omega t} \} = \text{Re} \{ |I| e^{j\theta_i} e^{j\omega t} \} = |I| \cos(\omega t + \theta_i) \quad \dots(ii)$$

Now, the instantaneous power is expressed as

$$\begin{aligned} \tilde{W} &= |V| |I| \cos(\omega t + \theta_v) \cos(\omega t + \theta_i) \\ &= \frac{|V| |I|}{2} [\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)] \quad \dots(iii) \end{aligned}$$

The right hand side of above equation consists of an average part and an oscillating part. If we define $\theta_v - \theta_i = \theta$ (see Fig. 10.6.1) we may write the average power as

$$W_{av} = \frac{|V| |I|}{2} \cos \theta \quad \dots(iv)$$

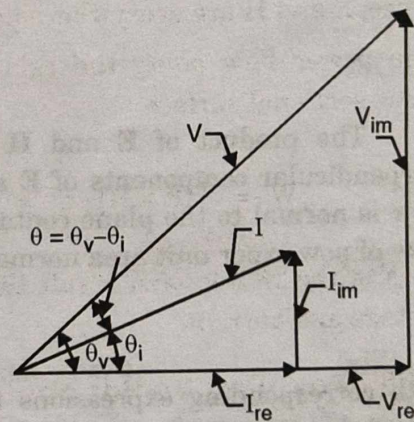


Fig. 10.6.1 : To find average and complex vector

Another useful quantity is the "reactive power" or more accurately, the reactive volt-amperes (VAR).

$$W_{react} = \frac{|V| |I|}{2} \sin \theta$$

Since, θ is the phase angle between the voltage and the current, W_{av} and W_{react} are the in-phase and out-of-phase components of the volt-amp product.

Consider now the complex power W , defined as one-half the product of V and the complex conjugate of I .



$$W = \frac{1}{2} VI^* = \frac{1}{2} |V| e^{j\theta_v} |I| e^{-j\theta_i} = \frac{|V||I|}{2} e^{j\theta} = W_{av} + jW_{react}$$

The average and reactive power may be recovered from the complex power by taking the real and imaginary parts.

$$W_{av} = \frac{1}{2} \operatorname{Re} \{VI^*\}$$

$$W_{react} = \frac{1}{2} \operatorname{Im} \{VI^*\}$$

In electromagnetic field theory there are relations similar to those between the Poynting vector, the electric field strength and the magnetic field strength. The instantaneous power flow per square

$$\tilde{\mathbf{P}} = \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}$$

...(A1)

Complex Poynting vector $\mathbf{P}$ can be expressed as

$$\mathbf{P} = \frac{1}{2} \mathbf{E} \times \mathbf{H}^*$$

...(A2)

from which we may obtain the average and reactive parts of the power flow per square meter.

$$P_{av} = \frac{1}{2} \operatorname{Re} \{ \mathbf{E} \times \mathbf{H}^* \}$$

...(A3)

$$P_{react} = \frac{1}{2} \operatorname{Im} \{ \mathbf{E} \times \mathbf{H}^* \}$$

...(A4)

The product of $\mathbf{E}$ and $\mathbf{H}$ in equation is a vector product. We know, mutually perpendicular components of $\mathbf{E}$ and $\mathbf{H}$ contribute to power flow and the direction of the flow is normal to the plane containing $\mathbf{E}$ and $\mathbf{H}$. In rectangular co-ordinates, the complex flow of power per unit area normal to the y-z plane is

$$P_x = \frac{1}{2} (E_y H_z^* - E_z H_y^*)$$

with corresponding expressions for the other directions. In spherical co-ordinates, the outward (radial) flow of complex power per unit area is

$$P_r = \frac{1}{2} (E_\theta H_\phi^* - E_\phi H_\theta^*)$$

10.6.1 Average Poynting Vector Expression :

As per the poynting theorem, the term $-\oint (\bar{\mathbf{E}} \times \bar{\mathbf{H}}) \cdot d\bar{\mathbf{s}}$ represents the rate of energy flow inward through the surface of volume. Here negative sign indicates inward flow. So the term $+\oint (\bar{\mathbf{E}} \times \bar{\mathbf{H}}) \cdot d\bar{\mathbf{s}}$ represents the energy flow in the outward direction through the surface enclosing the volume.



where η_m is the intrinsic impedance of the conductor inside the metallic conductor. The average (or real) power flow per unit area of the surface is

$$P_n(\text{real}) = \frac{1}{2} \text{Re}(\bar{E}_{\tan} \times \bar{H}_{\tan}^*)$$

When $\bar{E}_{\tan}$ and $\bar{H}_{\tan}$ are at right angles, and since for any good conductor $\bar{E}_{\tan}$ leads $\bar{H}_{\tan}$ by 45 degrees in time phase, becomes

$$\begin{aligned} P_n &= \frac{1}{2} |\bar{E}_{\tan}| |\bar{H}_{\tan}| \cos 45^\circ \\ &= \frac{1}{2\sqrt{2}} |\eta_m| |\bar{H}_{\tan}|^2 = \frac{1}{2\sqrt{2}} \frac{|\bar{E}_{\tan}|^2}{|\eta_m|} \end{aligned}$$

For a conductor which has a thickness very much greater than the skin depth δ , the surface impedance Z_s is always equal to the intrinsic impedance η_{in} of the conductor, so that

$$P_n = \frac{1}{2\sqrt{2}} |Z_s| |\bar{H}_{\tan}|^2 = \frac{1}{2\sqrt{2}} \frac{|\bar{E}_{\tan}|^2}{|Z_s|} \text{ watt/sq m}$$

In a conductor the linear current density J_s is equal in magnitude to the tangential magnetic field strength at the surface, so

$$P_n = \frac{1}{2\sqrt{2}} |Z_s| |J_s|^2 \text{ watt/sq m}$$

In expressions $\bar{E}_{\tan}$, $\bar{H}_{\tan}$ and J_s are peak values. In terms of effective values

$$\begin{aligned} P_n &= \frac{1}{\sqrt{2}} \frac{|\bar{E}_{t(\text{eff})}|^2}{|Z_s|} = \frac{1}{\sqrt{2}} |Z_s| |\bar{H}_{t(\text{eff})}|^2 \\ &= \frac{1}{\sqrt{2}} |Z_s| |J_{s(\text{eff})}|^2 = R_s J_{s(\text{eff})}^2 \text{ watt/sq m} \end{aligned}$$

10.7.1 Solved Examples on Poynting Theorem :

Important Formulae

$$\tilde{P} = \tilde{E} \times \tilde{H}$$

$$P = \frac{1}{2} E \times H^*$$

$$P_{\text{avg}} = \frac{1}{2} \text{Re} \{ E \times H^* \}$$

$$P_{\text{react}} = \frac{1}{2} \text{Im} \{ E \times H^* \}$$

$$P = \int_S \bar{P}_{\text{avg}} \cdot d\bar{s} \quad (\text{W})$$

$$P_{z, \text{avg}} = \frac{E_{x0}^2}{2\eta} \text{ s (Watts)}$$

Example 10.7.1 : The electric field intensity in a perfect dielectric medium is given as $\vec{E} = E_m \cos(\omega t - kz) \vec{a}_x$ (V/m), where E_m is the peak value and k is a constant. Determine the magnetic field intensity in the region and the direction of power flow.

Solution :

(i) From the given field expression is clear that the wave is travelling in + z direction. For this direction the E and H are related through the relation

$$\frac{E_x}{H_y} = -\frac{E_y}{H_x} = \eta$$

Since the given component of E is E_x , it will result in H_y .

$$H_y = \frac{E_x}{\eta} = \frac{E_m}{\eta} \cos(\omega t - kz)$$

or

$$\vec{H} = H_y \vec{a}_y = \frac{E_m}{\eta} \cos(\omega t - kz) \vec{a}_y \quad (\text{A/m})$$

Note that for perfect dielectric ($\sigma = 0$), the intrinsic impedance is real indicating E and H are in phase.

(ii) The direction of power flow is obtained using Poynting vector

$$\vec{P} = \vec{E} \times \vec{H} = E_x \vec{a}_x \times H_y \vec{a}_y = E_x H_y \vec{a}_z$$

Thus the power flow is in z-direction. It should be, because the wave is in + z direction.

Example 10.7.2 : A plane wave has electric field intensity E with $E_0 \cos(\omega t - kz) \vec{a}_y$ with $E_0 = 1000$ V/m and $f = 300$ MHz. For its propagation in free space (lossless)

(i) Calculate the Poynting vector

(ii) Instantaneous and time average power density in the wave.

Solution : From the given field expression the wave is travelling in + z direction and electric field has E_y component. From free space properties.

$$v = c = 3 \times 10^8 \text{ m/s}, \quad \text{and} \quad \eta = 120\pi$$

we calculate phase constant k

$$k = \frac{\omega}{c} = \frac{2\pi \times 300 \times 10^6}{3 \times 10^8} = 2\pi \text{ (rad/m)}$$

To find the Poynting vector we require $\vec{H}$. In general form

$$\vec{H} = H_0 \cos(\omega t - kz) \vec{a}_H$$

where

$$H_0 = \frac{E_0}{\eta} = \frac{1000}{120\pi} = 2.65 \text{ (A/m)}.$$

and

$$\vec{a}_E \times \vec{a}_H = \vec{a}_z$$

i.e $\bar{a}_y \times \bar{a}_H = \bar{a}_z$

or $\bar{a}_H = -\bar{a}_x$

then $\bar{H} = -2.65 \cos(\omega t - kz) \bar{a}_x \text{ (A/m)}$

...Ans.

(i) To find the Poynting vector :

$$\begin{aligned}\bar{P} &= \bar{E} \times \bar{H} \\ &= 1000 \cos(\omega t - kz) \bar{a}_y \times -2.65 \cos(\omega t - kz) \bar{a}_x \\ &= +2650 \cos^2(\omega t - kz) \bar{a}_z \\ &= 2650 \left[\frac{1 + \cos 2(\omega t - kz)}{2} \right] \bar{a}_z \\ &= 1325 [1 + \cos(2\omega t - 2kz)] \bar{a}_z \text{ (W/m}^2\text{)}\end{aligned}$$

...Ans.

(ii) Instantaneous and time average power density :

The Poynting vector itself gives the instantaneous power density. So it is

$$\bar{P} = 1325 [1 + \cos(2\omega t - 2kz)] \bar{a}_z \text{ (W/m}^2\text{)}$$

To obtain average power density

$$\bar{P}_{av} = \frac{1}{2\eta} E_0^2 \bar{a}_z = \frac{1}{2 \times 120\pi} \times 1000^2 \bar{a}_z = 1325 \bar{a}_z \text{ (W/m}^2\text{)} \quad \text{...Ans.}$$

You can see this value in the $\bar{P}$ as a dc term.

Example 10.7.3 : A linearly polarized plane wave propagates through free space at an angle θ with respect to the z-axis. If the peak amplitude of the electric field intensity $\bar{E}$ is 10 (mV/m) calculate the average power that passes through a surface of area 1 m² in the z = 0 plane.

Solution : Electric field intensity $E_m = 10 \text{ mV/m}$, Surface area = 1 m².

As the wave is propagating in free space, the intrinsic impedance of the free space is $\eta_0 = 377 \Omega$.

$\therefore$ The average Poynting vector is given by

$$P_{avg} = \frac{1}{2} \cdot \frac{E_{x0}^2}{\eta} = \frac{1}{2} \cdot \frac{E_m^2}{\eta_0} = \frac{1}{2} \times \frac{(10 \times 10^{-3})^2}{377} = 0.0013 \times 10^{-4}$$

$$\therefore P_{avg} = 0.13 \mu\text{Watt/m}^2$$

The given area is in z = 0 plane for which the normal is in z-direction. But the wave is not in the same direction, it is making angle θ with z-direction. Projecting P_{avg} in z-direction.

$$P_{z, avg} = P_{avg} \cos \theta = 0.13 \cos \theta \text{ (}\mu\text{W/m}^2\text{)}$$

So, the power crossing 1 m^2 area is

$$P = 0.13 \cos \theta \times (1) = 0.13 \cos \theta \text{ (W)}$$

...Ans.

Example 10.7.4 : $E(z, t) = 50 \cos (\omega t - \beta z) \bar{a}_x$ V/m in free space. Find the average power crossing the circular area of radius 2.5 m in free space with z constant.

Solution : From the given field expression the wave is travelling in $+z$ -direction. The average power in this wave is

$$P_{z, \text{avg}} = \frac{1}{2} \frac{E_{x0}^2}{\eta}$$

for free space $\eta = 120 \pi$ and given $E_{x0} = 50 \text{ V/m}$

$$P_{z, \text{avg}} = \frac{1}{2} \times \frac{50^2}{120 \pi} = 3.32 \text{ (W/m}^2\text{)} = P_{\text{avg}}$$

The given wave is travelling in z -direction which is perpendicular to the given $z = \text{constant}$ surface. So the power is given by, Equation (10.6.4)

$$P = P_{\text{avg}} S = 3.32 \times \pi (2.5)^2 = 65.2 \text{ (W)}$$

...Ans.

Example 10.7.5 : Consider a circular cylinder with radius of one meter and length 0.75 m in free space with its axis along the z -direction. An EM wave is propagating in the positive z -direction with its electric field

$$\bar{E} = \cos \left[2\pi f \left(t - \frac{z}{c} \right) \right] \bar{a}_x \text{ (V/m)}$$

where, $f = 100 \text{ MHz}$ and ' c ' is the velocity of light.

Determine : (i) Poynting vector, (ii) net power flow entering the cylinder.

Solution : (i) To find Poynting vector :

The given expression can be written in standard form as

$$\bar{E} = \cos \left[2\pi f t - \frac{2\pi f z}{c} \right] \bar{a}_x = \cos (\omega t - \beta z) \bar{a}_x$$

where, $\beta = 2\pi f / c$

The magnetic field related to given $\bar{E}$ as

$$\bar{H} = \frac{\bar{E}}{\eta} = \frac{1}{120\pi} \cos (\omega t - \beta z) \bar{a}_y \text{ (note } \bar{a}_y \text{ direction)}$$

The Poynting vector is

$$\bar{P} = \bar{E} \times \bar{H} = (1) \left(\frac{1}{120\pi} \right) \cos^2 (\omega t - \beta z) \bar{a}_z$$

The average power in z -direction is

$$P_{z, \text{avg}} = \frac{1}{2} \frac{E_{x0}^2}{\eta} = \frac{1}{2} \times \frac{1^2}{120\pi} = 1.326 \times 10^{-3} \text{ (W/m}^2\text{)} = P_{\text{avg}}$$

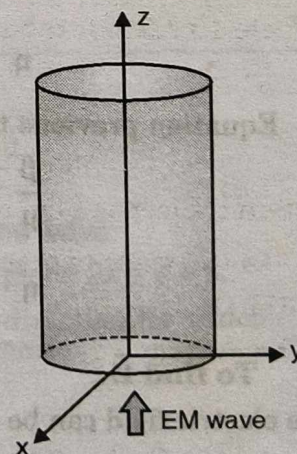


Fig. : Illustrating Ex. 10.7.5

ii) The circular area is normal to the wave direction, thus the net power flow entering the cylinder is



$$P = P_{avg} \times \text{area}$$

$$= 1.326 \times 10^{-3} \times \pi \times (1)^2 = 4.17 \text{ mW}$$

...Ans.

Example 10.7.6 : In a non magnetic material ($\epsilon_r \neq 0$, $\mu = \mu_0$, $\sigma = 0$)

- Find E using Maxwell equations
- Find poynting vector
- Find the time average power crossing the surface $x = 1$, $0 < y < 2$, $0 < z < 3$ m.

$$\text{given, } \bar{H} = 30 \cos (2\pi \times 10^8 t - 6x) \bar{a}_y \text{ (mA/m)}$$

Solution :

$$\text{Given : } \bar{H} = 30 \cos (2\pi \times 10^8 t - 6x) \bar{a}_y = H_0 \cos (2\pi \times 10^8 t - 6x) \bar{a}_H \left(\frac{\text{mA}}{\text{m}} \right)$$

From which we get,

$$\omega = 2\pi \times 10^8 \text{ (rad/s),}$$

$$\beta = 6$$

and the wave is travelling in + x direction, i.e. $\bar{a}_\beta = \bar{a}_x$. To obtain electric field we require intrinsic impedance it is obtained as follows :

$$\beta = \omega \sqrt{\mu \epsilon}$$

dividing by μ

$$\frac{\beta}{\mu} = \omega \sqrt{\frac{\epsilon}{\mu}}$$

but for a lossless media ($\because \sigma = 0$), the intrinsic impedance

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

$\therefore$ Equation previous to above equation becomes

$$\frac{\beta}{\mu} = \frac{\omega}{\eta}$$

or

$$\eta = \frac{\omega \mu}{\beta} = \frac{2\pi \times 10^8 \times 4\pi \times 10^{-7}}{6} = 131.59 \text{ } (\Omega)$$

...Ans.

i) **To find $\bar{E}$:**

The electric field can be written as,

$$\bar{E} = E_0 \cos (2\pi \times 10^8 t - 6x) \bar{a}_E$$

The amplitude E_0 is obtained using,

$$\eta = \frac{E_0}{H_0}$$

$$E_0 = \eta H_0 = 131.59 \times 30 \times 10^{-3} = 3.948$$

To obtain $\vec{a}_E$,

$$\vec{a}_E \times \vec{a}_H = \vec{a}_\beta$$

$$\vec{a}_E \times \vec{a}_y = \vec{a}_x$$

$$\vec{a}_E = -\vec{a}_z$$

$$\vec{E} = -3.948 \cos(2\pi \times 10^8 t - 6x) \vec{a}_z \text{ (V/m)}$$

...Ans.

ii) To find Poynting vector :

$$\vec{P} = \vec{E} \times \vec{H}$$

$$= -3.948 \cos(2\pi \times 10^8 t - 6x) \vec{a}_z \times 30 \cos(2\pi \times 10^8 t - 6x) \vec{a}_y \times (10^{-3})$$

$$= 0.1184 \cos^2(2\pi \times 10^8 t - 6x) \vec{a}_x \text{ (W/m}^2\text{)}$$

...Ans.

iii) To find average power :

Average power density is

$$\vec{P}_{\text{avg}} = \frac{1}{2} \eta H_0^2 \vec{a}_x = \frac{1}{2} \times 131.59 \times (30 \times 10^{-3})^2 = 0.0592 \vec{a}_x \text{ (W/m}^2\text{)}$$

Average power crossing the given area in $x = 1$ plane is,

$$P = \int_S \vec{P}_{\text{avg}} \cdot d\vec{s}$$

$$= \int_S 0.0592 \vec{a}_x \cdot dy dz \vec{a}_x = \int_0^3 \int_0^2 0.0592 dy dz$$

$$= 0.0592 \times 2 \times 3 = 0.3535 \text{ (W)}$$

...Ans.

Example 10.7.7 : In a non magnetic medium

$$\vec{E} = 4 \sin(2\pi \times 10^7 t - 0.8x) \vec{a}_z \text{ (V/m)}$$

Find, (i) ϵ_r, η

(ii) The time average power carried by the wave

(iii) The total power crossing 100 cm^2 of plane $2x + y = 5$.

Solution : From the given field expression, the wave is in $+x$ direction for which,

$$E_0 = 4, \quad \omega = 2\pi \times 10^7, \quad \beta = 0.8$$

The velocity of the wave in this media is

$$v = \frac{\omega}{\beta} = \frac{2\pi \times 10^7}{0.8} = 0.78 \times 10^8 \text{ (m/s)} \neq c$$

It implies that the medium is not a free space.

(i) To find ϵ_r, η :

$$\beta = \omega \sqrt{\mu \epsilon} = \omega \sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r} = \frac{\omega}{c} \sqrt{\epsilon_r}$$